

对线性方程组, 令

$$\alpha_j = \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix} \quad (j=1, 2, \dots, n), \quad \beta = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

则线性方程组可表示为如下向量形式:

$$x_1 \alpha_1 + x_2 \alpha_2 + \dots + x_n \alpha_n = \beta$$

线性方程组是否有解 $\Leftrightarrow$ 是否存在一组数 $k_1, k_2, \dots, k_n$, 使 $\beta = k_1 \alpha_1 + k_2 \alpha_2 + \dots + k_n \alpha_n$ 这一线性关系式成立.

例如, 对 $\alpha_1 = (1, 2, -1)$, $\alpha_2 = (2, 3, -1)$, $\alpha_3 = (4, 1, -1)$, 有 $\alpha_3 = 2\alpha_1 + \alpha_2$

即 α_3 是 α_1, α_2 的线性组合, 或称 α_3 可由 α_1, α_2 线性表示

$k_1 \alpha_1 + k_2 \alpha_2 + \dots + k_n \alpha_n = \beta$ 有唯一解 $\Leftrightarrow \beta$ 可由向量组 $\alpha_1, \alpha_2, \dots, \alpha_n$ 唯一线性表示

$k_1 \alpha_1 + k_2 \alpha_2 + \dots + k_n \alpha_n = \beta$ 有无穷多组解 $\Leftrightarrow \beta$ 可由向量组 $\alpha_1, \alpha_2, \dots, \alpha_n$ 线性表示且表示不唯一

$k_1 \alpha_1 + k_2 \alpha_2 + \dots + k_n \alpha_n = \beta$ 无解 $\Leftrightarrow \beta$ 不可由向量组 $\alpha_1, \alpha_2, \dots, \alpha_n$ 线性表示

向量 β 可由向量组 $\alpha_1, \alpha_2, \dots, \alpha_n$ 线性表示 $\Leftrightarrow A = (\alpha_1, \alpha_2, \dots, \alpha_n)$ 的秩等于 $B = (\alpha_1, \alpha_2, \dots, \alpha_n, \beta)$ 的秩

例: 设有向量组 $\alpha_1 = \begin{bmatrix} a \\ 2 \\ 10 \end{bmatrix}$, $\alpha_2 = \begin{bmatrix} -2 \\ 1 \\ 5 \end{bmatrix}$, $\alpha_3 = \begin{bmatrix} -1 \\ 1 \\ 4 \end{bmatrix}$, $\beta = \begin{bmatrix} 1 \\ b \\ c \end{bmatrix}$.

试问当 a, b, c 满足什么条件时:

(1) β 可由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示且表示方法唯一

$a_1 x_1 + a_2 x_2 + a_3 x_3 = \beta$ 有唯一解, 即 $\begin{bmatrix} a & -2 & -1 \\ 2 & 1 & 1 \\ 10 & 5 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ b \\ c \end{bmatrix}$ 有唯一解

$\begin{vmatrix} a & -2 & -1 \\ 2 & 1 & 1 \\ 10 & 5 & 4 \end{vmatrix} \xrightarrow{r_3 - 5r_2} \begin{vmatrix} a & -2 & -1 \\ 2 & 1 & 1 \\ 0 & 0 & -1 \end{vmatrix} = (-1) \times (-1)^{3+2} \times (a+4) = -a-4$, 要使 $\begin{bmatrix} a & -2 & -1 \\ 2 & 1 & 1 \\ 10 & 5 & 4 \end{bmatrix}$ 可逆, 则 $-a-4 \neq 0$ 即 $a \neq -4$

(2) β 不能由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示

即 $B = \begin{bmatrix} a & -2 & -1 & 1 \\ 2 & 1 & 1 & b \\ 10 & 5 & 4 & c \end{bmatrix} \xrightarrow{\substack{2r_2+r_1 \\ 13+2r_1-r_2}} \begin{bmatrix} a & -2 & -1 & 1 \\ 4+a & 0 & 1 & 2b+1 \\ 8+a & 0 & 1 & c+2-b \end{bmatrix} \xrightarrow{13-r_2} \begin{bmatrix} a & -2 & -1 & 1 \\ 4+a & 0 & 1 & 2b+1 \\ 4+a & 0 & 0 & c+1-3b \end{bmatrix}$

$R(A) \neq R(B)$
3 2 1

当 $a = -4$ 时, $c+1-3b \neq 0$ 即可满足 $R(A) \neq R(B)$.

即 $(a = -4 \text{ 且 } 3b - c \neq 1)$

(3) β 可由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示, 但表示方法不唯一, 写出一般表达式

即要使 $R(A) = R(B) < 3$, 故 $a = -4$ 且 $c+1-3b = 0$

即 $(a = -4 \text{ 且 } 3b - c = 1)$

$B \rightarrow \begin{bmatrix} -4 & -2 & -1 & 1 \\ 0 & 0 & 1 & 2b+1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\substack{r_1+r_2 \\ r_1 \times (-\frac{1}{2})}} \begin{bmatrix} 1 & \frac{1}{2} & 0 & -\frac{1}{2}b - \frac{1}{2} \\ 0 & 0 & 1 & 2b+1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, 得 $\begin{cases} 2x_1 + x_2 = -b-1 \\ x_3 = 2b+1 \end{cases}$

令 $x_2 = c (c \in \mathbb{R})$ 则 $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = c \begin{bmatrix} -\frac{1}{2} \\ 1 \\ 2b+1 \end{bmatrix} + \begin{bmatrix} -\frac{1}{2}b - \frac{1}{2} \\ 0 \\ 2b+1 \end{bmatrix}$

所以表达式为 $\beta = (-\frac{1}{2}c - \frac{1}{2}b - \frac{1}{2})\alpha_1 + c\alpha_2 + (2b+1)\alpha_3$, c 为任意实数

例 设 $\alpha_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 2 \end{bmatrix}$, $\alpha_2 = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix}$, $\alpha_3 = \begin{bmatrix} 1 \\ -1 \\ 4 \\ 0 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 0 \\ 3 \\ 1 \end{bmatrix}$, 证明向量 b 能由向量组 $\alpha_1, \alpha_2, \alpha_3$ 线性表示, 并求出表示式

证明:

$$B = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & -1 & 0 \\ 2 & 1 & 4 & 3 \\ 2 & 3 & 0 & 1 \end{bmatrix} \xrightarrow[r_4 - r_1]{\substack{r_2 - r_1 \\ r_3 - 2r_1}} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & -2 & -1 \\ 0 & -1 & 2 & 1 \\ 0 & 1 & -2 & -1 \end{bmatrix} \xrightarrow[r_4 - r_2]{r_3 + r_2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$r_1 - r_2 \rightarrow \begin{bmatrix} 1 & 0 & 3 & 2 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

故 $R(A) = R(B)$.

$\therefore$ 向量 b 能由向量组 $\alpha_1, \alpha_2, \alpha_3$ 线性表示.

$$\begin{cases} x_1 + 3x_3 = 2 \\ x_2 - 2x_3 = -1 \end{cases}$$

$(\alpha_1, \alpha_2, \alpha_3) X = b$ 的通解为

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = C \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -3C + 2 \\ 2C - 1 \\ C \end{bmatrix}$$

令 $x_3 = C (C \in \mathbb{R})$

$$\therefore \begin{cases} x_1 = 2 - 3C \\ x_2 = 2C - 1 \\ x_3 = C \end{cases}$$

$$\therefore b = (\alpha_1, \alpha_2, \alpha_3) X = (\alpha_1, \alpha_2, \alpha_3) \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = (-3C + 2)\alpha_1 + (2C - 1)\alpha_2 + C\alpha_3 (C \in \mathbb{R})$$

设有两个向量组 $A: \alpha_1, \alpha_2, \dots, \alpha_r$ 和 $B: \beta_1, \beta_2, \dots, \beta_s$, 若 B 中每一个向量均可由 A 中的向量线性表示, 则称向量组 B 可由向量组 A 线性表示

若向量组 A 和向量组 B 可以互相线性表示, 则称这两个向量组等价

若向量组 B 可由向量组 A 线性表示, 则 $(\alpha_1, \alpha_2, \dots, \alpha_r) X = (\beta_1, \beta_2, \dots, \beta_s)$

向量组 $B: \beta_1, \beta_2, \dots, \beta_s$ 能由向量组 $A: \alpha_1, \alpha_2, \dots, \alpha_r$ 线性表示的充要条件:

矩阵 $A = (\alpha_1, \alpha_2, \dots, \alpha_r)$ 的秩等于矩阵 $(A, B) = (\alpha_1, \alpha_2, \dots, \alpha_r, \beta_1, \beta_2, \dots, \beta_s)$ 的秩, 即 $R(A) = R(A, B)$

向量组 $B: \beta_1, \beta_2, \dots, \beta_s$ 与向量组 $A: \alpha_1, \alpha_2, \dots, \alpha_r$ 等价的充要条件 $R(A) = R(B) = R(A, B)$

例. 设 $a_1 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ 1 \end{bmatrix}$, $a_2 = \begin{bmatrix} 3 \\ 1 \\ 1 \\ 3 \end{bmatrix}$, $b_1 = \begin{bmatrix} 2 \\ 0 \\ 1 \\ 1 \end{bmatrix}$, $b_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 2 \end{bmatrix}$, $b_3 = \begin{bmatrix} 3 \\ -1 \\ 2 \\ 0 \end{bmatrix}$,

证明向量组 a_1, a_2 与向量组 b_1, b_2, b_3 等价.

证明: 记 $A = (a_1, a_2)$, $B = (b_1, b_2, b_3)$

$$(A, B) = \begin{bmatrix} 1 & 3 & 2 & 1 & 3 \\ -1 & 1 & 0 & 1 & -1 \\ 1 & 1 & 1 & 0 & 2 \\ -1 & 3 & 1 & 2 & 0 \end{bmatrix} \xrightarrow[r_4 + r_1]{\substack{r_2 + r_1 \\ r_3 - r_1}} \begin{bmatrix} 1 & 3 & 2 & 1 & 3 \\ 0 & 4 & 2 & 2 & 2 \\ 0 & -2 & -1 & -1 & -1 \\ 0 & 6 & 3 & 3 & 3 \end{bmatrix} \xrightarrow{r_4 + 3r_3} \begin{bmatrix} 1 & 3 & 2 & 1 & 3 \\ 0 & 4 & 2 & 2 & 2 \\ 0 & -2 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[r_2 \times \frac{1}{2}]{r_3 + \frac{1}{2}r_2} \begin{bmatrix} 1 & 3 & 2 & 1 & 3 \\ 0 & 2 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

可见, $R(A) = R(B) = R(A, B) = 2$, 故二者等价

若向量组 $B: \beta_1, \beta_2, \dots, \beta_s$ 可由向量组 $A: \alpha_1, \alpha_2, \dots, \alpha_r$ 线性表示, 则 $R(\beta_1, \beta_2, \dots, \beta_s) \leq R(\alpha_1, \alpha_2, \dots, \alpha_r)$

若向量组 $\alpha_1, \dots, \alpha_m$ 与向量组 $b_1, \dots, b_s$ 等价,

则它们的向量空间 $V_1 = V_2$

对于向量组 $\alpha_1, \alpha_2, \dots, \alpha_m$, 若有一组不全为零的数 $k_1, k_2, \dots, k_m$, 使

$$k_1\alpha_1 + k_2\alpha_2 + \dots + k_m\alpha_m = 0$$

则称 $\alpha_1, \alpha_2, \dots, \alpha_m$ 线性相关

若仅当 $k_1 = k_2 = \dots = k_m = 0$ 时, $k_1\alpha_1 + k_2\alpha_2 + \dots + k_m\alpha_m = 0$ 才成立, 则称 $\alpha_1, \alpha_2, \dots, \alpha_m$ 线性无关

例. 向量组 $\alpha_1, \alpha_2, \alpha_3$ 线性无关, $\beta_1 = \alpha_1 + \alpha_2$, $\beta_2 = \alpha_2 + \alpha_3$, $\beta_3 = \alpha_3 + \alpha_1$, 讨论向量组 $\beta_1, \beta_2, \beta_3$ 的线性相关性

解: 设 $k_1\beta_1 + k_2\beta_2 + k_3\beta_3 = 0$

$$\text{即 } k_1(\alpha_1 + \alpha_2) + k_2(\alpha_2 + \alpha_3) + k_3(\alpha_3 + \alpha_1) = 0$$

$$(k_1 + k_3)\alpha_1 + (k_1 + k_2)\alpha_2 + (k_2 + k_3)\alpha_3 = 0$$

由于 $\alpha_1, \alpha_2, \alpha_3$ 线性无关, 得 $\begin{cases} k_1 + k_3 = 0 \\ k_1 + k_2 = 0 \\ k_2 + k_3 = 0 \end{cases}$

☆ 其系数行列式 $\begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = 2 \neq 0$, 即齐次线性方程组仅有零解

即: 仅当 $k_1 = k_2 = k_3 = 0$ 时, $k_1\beta_1 + k_2\beta_2 + k_3\beta_3 = 0$ 才成立, 故 $\beta_1, \beta_2, \beta_3$ 线性无关

向量组 $\alpha_1, \alpha_2, \dots, \alpha_m$ ($m \geq 2$) 线性相关的充要条件是其中至少有 1 个向量可由其余 $(m-1)$ 个向量线性表示.

逆否

向量组 $\alpha_1, \alpha_2, \dots, \alpha_m$ 线性无关的充要条件是其中任一向量均不可由其余向量线性表示

零向量可以由任何向量线性表示, 含有零向量的任何向量组必线性相关

向量组 $\alpha_1, \alpha_2, \dots, \alpha_m$ 线性相关的充要条件是矩阵 $A = (\alpha_1, \alpha_2, \dots, \alpha_m)$ 的秩小于向量个数 m ,
线性无关的充要条件是 $R(A) = m$ 即 $R(A) < m$

① 若向量组 $\alpha_1, \alpha_2, \dots, \alpha_r$ 线性相关, 则向量组 $\alpha_1, \alpha_2, \dots, \alpha_r, \alpha_{r+1}, \dots, \alpha_m$ 也线性相关 部分相关, 全体相关

若 $\alpha_1, \alpha_2, \dots, \alpha_r, \alpha_{r+1}, \dots, \alpha_m$ 线性无关, 则向量组 $\alpha_1, \alpha_2, \dots, \alpha_r$ 也线性无关 全体无关, 部分无关

② 对 m 个 n 维向量组成的向量组, 当向量个数 m 大于向量维数 n 时, 向量组必线性相关

③ 设向量组 $\alpha_1, \alpha_2, \dots, \alpha_m$ 线性无关, 而向量组 $\alpha_1, \alpha_2, \dots, \alpha_m, \beta$ 线性相关,

则 β 可由 $\alpha_1, \alpha_2, \dots, \alpha_m$ 线性表示且表示式唯一

例. 讨论下列向量组的线性相关性

$$(1) \alpha_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \alpha_2 = \begin{bmatrix} 0 \\ 3 \\ 2 \end{bmatrix}, \alpha_3 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$$

$$\text{解: } (\alpha_1, \alpha_2, \alpha_3) = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 3 & 2 \\ 1 & 2 & 1 \end{bmatrix} \xrightarrow[r_3-r_1]{r_2-r_1} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 3 & 3 \\ 0 & 2 & 2 \end{bmatrix}$$

$$\xrightarrow[r_3-2r_2]{r_2 \times \frac{1}{3}} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$R(\alpha_1, \alpha_2, \alpha_3) = 2 < m = 3$ 故 $\alpha_1, \alpha_2, \alpha_3$ 线性相关

$$(2) \alpha_1 = \begin{bmatrix} 5 \\ 2 \\ 9 \\ 1 \end{bmatrix}, \alpha_2 = \begin{bmatrix} 2 \\ 1 \\ 2 \\ 3 \end{bmatrix}, \alpha_3 = \begin{bmatrix} 7 \\ 13 \\ 11 \\ 4 \end{bmatrix}$$

$$\text{解: } (\alpha_1, \alpha_2, \alpha_3) = \begin{bmatrix} 5 & 2 & 7 \\ 2 & 1 & 13 \\ 9 & 2 & 11 \\ 1 & 3 & 4 \end{bmatrix} \xrightarrow[r_3-9r_4]{r_1-5r_4, r_2-2r_4} \begin{bmatrix} 0 & -13 & -13 \\ 0 & -5 & 5 \\ 0 & -25 & -25 \\ 1 & 3 & 4 \end{bmatrix}$$

$$\xrightarrow[r_3 \times (-\frac{1}{5})]{r_2 \times (-\frac{1}{5})} \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \\ 1 & 3 & 4 \end{bmatrix} \xrightarrow[r_3-r_2]{r_1 \leftrightarrow r_2} \begin{bmatrix} 1 & 3 & 4 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{r_4-r_3} \begin{bmatrix} 1 & 3 & 4 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow[r_3-r_2]{r_1-3r_2} \begin{bmatrix} 1 & 3 & 4 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{r_3 \times \frac{1}{2}} \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$R(\alpha_1, \alpha_2, \alpha_3) = 3 = m$ 故 $\alpha_1, \alpha_2, \alpha_3$ 线性无关

例 设向量组 $\alpha_1, \alpha_2, \alpha_3$ 线性相关, 向量组 $\alpha_2, \alpha_3, \alpha_4$ 线性无关, 证明:

(1) α_1 能由 α_2, α_3 线性表示

(2) α_4 不能由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示

证明: $\because \alpha_2, \alpha_3, \alpha_4$ 线性无关

$\therefore \alpha_2, \alpha_3$ 线性无关

$\because \alpha_1, \alpha_2, \alpha_3$ 线性相关

$\therefore \alpha_1$ 可由 α_2, α_3 线性表示

证明: 假设 α_4 能由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示

$\because \alpha_1$ 能由 α_2, α_3 线性表示

$\therefore \alpha_4$ 能由 α_2, α_3 线性表示

与 $\alpha_2, \alpha_3, \alpha_4$ 线性无关矛盾

α_4 不能由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示

$$\text{设 } \alpha_j = \begin{bmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{bmatrix}, b_j = \begin{bmatrix} a_{1j} \\ \vdots \\ a_{mj} \\ a_{m+1,j} \\ \vdots \\ a_{nj} \end{bmatrix} \quad (j=1, 2, \dots, m)$$

即向量 α_j 添加若干分量后变为向量 b_j .

若向量组 $\alpha_1, \alpha_2, \dots, \alpha_m$ 线性无关, 则向量组 $b_1, b_2, \dots, b_m$ 也线性无关 **低维无关, 高维无关**

反之,

若向量组 $b_1, b_2, \dots, b_m$ 线性相关, 则向量组 $\alpha_1, \alpha_2, \dots, \alpha_m$ 也线性相关 **高维相关, 低维相关**

小练习

Number 10

1. 试问下列向量 β 能否由其余向量线性表示? 若能, 写出其线性表示式.

(1) $\alpha_1 = (1, 2)^T$, $\alpha_2 = (-1, 0)^T$, $\beta = (3, 4)^T$

(2) $\alpha_1 = (1, 0, 2)^T$, $\alpha_2 = (2, -8, 0)^T$, $\beta = (1, 2, -1)^T$

解:

$$\therefore \alpha_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \alpha_2 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \beta = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$\text{令 } A = (\alpha_1, \alpha_2) = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix} \xrightarrow{r_2 - 2r_1} \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$$

则 $R(A) = 2$

$$(A, \beta) = \begin{bmatrix} 1 & -1 & 3 \\ 2 & 0 & 4 \end{bmatrix} \xrightarrow{r_2 - 2r_1} \begin{bmatrix} 1 & -1 & 3 \\ 0 & 2 & -2 \end{bmatrix}, \text{ 则 } R(A, \beta) = 2$$

$R(A) = R(A, \beta)$, 可线性表示且 $\beta = 2\alpha_1 - \alpha_2$

2. 已知向量组 $A: \alpha_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \alpha_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}; \beta: \beta_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \beta_2 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \beta_3 = \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}$

证明向量组 A 与向量组 B 等价

解:

$$\text{记 } A = (\alpha_1, \alpha_2) = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\xrightarrow{r_1 \leftrightarrow r_2} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \xrightarrow{r_3 - r_1} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & -1 \end{bmatrix} \xrightarrow{r_3 + r_2} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \text{ 则 } R(A) = 2$$

$$\text{记 } B = (\beta_1, \beta_2, \beta_3) = \begin{bmatrix} -1 & 1 & 3 \\ 0 & 2 & 2 \\ 1 & 1 & -1 \end{bmatrix} \xrightarrow{r_3 + r_1} \begin{bmatrix} -1 & 1 & 3 \\ 0 & 2 & 2 \\ 0 & 2 & 2 \end{bmatrix} \xrightarrow{r_3 - r_2} \begin{bmatrix} -1 & 1 & 3 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \end{bmatrix} \text{ 则 } R(B) = 2$$

$$\text{而 } (A, B) = \begin{bmatrix} 0 & 1 & -1 & 1 & 3 \\ 1 & 1 & 0 & 2 & 2 \\ 1 & 0 & 1 & 1 & -1 \end{bmatrix} \xrightarrow{r_2 - r_3} \begin{bmatrix} 0 & 1 & -1 & 1 & 3 \\ 0 & 1 & -1 & 1 & 3 \\ 1 & 0 & 1 & 1 & -1 \end{bmatrix} \xrightarrow{r_1 - r_2} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 1 & 3 \\ 1 & 0 & 1 & 1 & -1 \end{bmatrix} \xrightarrow{r_1 \leftrightarrow r_3} \begin{bmatrix} 1 & 0 & 1 & 1 & -1 \\ 0 & 1 & -1 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \text{ 则 } R(A, B) = 2$$

$\therefore R(A) = R(B) = R(A, B) = 2$, 故二者等价

3. 已知 $R(\alpha_1, \alpha_2, \alpha_3) = 2$, $R(\alpha_2, \alpha_3, \alpha_4) = 3$, 证明:

(1) α_1 能由 α_2, α_3 线性表示

证明:

$$\therefore R(\alpha_2, \alpha_3, \alpha_4) = 3$$

$\therefore \alpha_2, \alpha_3, \alpha_4$ 线性无关

从而 α_2, α_3 线性无关

$$\therefore R(\alpha_1, \alpha_2, \alpha_3) = 2 < 3$$

$\therefore \alpha_1, \alpha_2, \alpha_3$ 线性相关

故 α_1 能被 α_2, α_3 线性表示

(2) α_4 不能由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示

证明:

假设 α_4 能由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示.

由于 α_1 能由 α_2, α_3 线性表示

则 α_4 能由 α_2, α_3 线性表示.

这与 $R(\alpha_2, \alpha_3, \alpha_4) = 3 \Leftrightarrow \alpha_2, \alpha_3, \alpha_4$ 线性无关

即 α_4 不可由 α_2, α_3 线性表示矛盾

故 α_4 不能由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示

假设不成立,

4. 设有向量 $\alpha_1 = \begin{bmatrix} 1+\lambda \\ 1 \\ 1 \end{bmatrix}$, $\alpha_2 = \begin{bmatrix} 1 \\ 1+\lambda \\ 1 \end{bmatrix}$, $\alpha_3 = \begin{bmatrix} 1 \\ 1 \\ 1+\lambda \end{bmatrix}$, $\beta = \begin{bmatrix} 0 \\ \lambda \\ \lambda^2 \end{bmatrix}$, 试问当 λ 取何值时:

求具体表达式呢?

(1) β 可由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示, 且表达式唯一? 方法二 $|\alpha_1, \alpha_2, \alpha_3| = \lambda^2(\lambda+3)$

解: 设有一组数 x_1, x_2, x_3 , 使 $(\alpha_1, \alpha_2, \alpha_3) \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \beta = \begin{bmatrix} 0 \\ \lambda \\ \lambda^2 \end{bmatrix}$

则有 $\begin{cases} (1+\lambda)x_1 + x_2 + x_3 = 0 \\ x_1 + (1+\lambda)x_2 + x_3 = \lambda \\ x_1 + x_2 + (1+\lambda)x_3 = \lambda^2 \end{cases}$

增广矩阵 $(A, \beta) = \begin{bmatrix} 1+\lambda & 1 & 1 & 0 \\ 1 & 1+\lambda & 1 & \lambda \\ 1 & 1 & 1+\lambda & \lambda^2 \end{bmatrix} \xrightarrow{r_2-r_1, r_3-r_1} \begin{bmatrix} 1+\lambda & 1 & 1 & 0 \\ 0 & \lambda & -\lambda & \lambda(1-\lambda) \\ 0 & 0 & -\lambda(\lambda+3) & -\lambda^3+\lambda^2+\lambda \end{bmatrix}$

要使 $R(A) = R(A, \beta) = 3$, 则 $\lambda \neq 0$ 且 $\lambda \neq -3$

当 $\lambda \neq 0$ 且 $\lambda \neq -3$ 时, 只有零解, 即有唯一解, 可唯一线性表示

当 $\lambda = 0$ 时, 有无穷多解, 可线性表示但表达式不唯一

当 $\lambda = -3$ 时, 系数矩阵与增广矩阵 A 秩不相等, 无解

(2) β 可由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示, 但表达式不唯一?

解: 要使 $R(A) = R(A, \beta) < n$

① 若 $R(A) = R(A, \beta) = 1$, 令 $R(A) = 1$, 则 $\lambda = 0$, 满足 $R(A, \beta) = 1$

故 $\lambda = 0$

② 若 $R(A) = R(A, \beta) = 2$, 令 $R(A) = 2$, 则 $\lambda = -3$, 不满足 $R(A, \beta) = 2$.

故不成立.

(3) β 不能由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示?

解: 要使 $R(A) < R(A, \beta)$

① 若 $R(A) = 1$, 则 $\lambda = 0$, $(A, \beta) \rightarrow \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, 则 $R(A, \beta) = 0$ 不成立

② 若 $R(A) = 2$, 则 $\lambda = -3$, $(A, \beta) \rightarrow \begin{bmatrix} 1 & 1 & -2 & 9 \\ 0 & -3 & 3 & -12 \\ 0 & 0 & 0 & 6 \end{bmatrix}$, 则 $R(A, \beta) = 3$. 满足

故 $\lambda = -3$

5. 设有向量 $\alpha_1 = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 3 \end{bmatrix}$, $\alpha_2 = \begin{bmatrix} 1 \\ 1 \\ 3 \\ 5 \end{bmatrix}$, $\alpha_3 = \begin{bmatrix} 1 \\ -1 \\ a+2 \\ 1 \end{bmatrix}$, $\alpha_4 = \begin{bmatrix} 1 \\ 2 \\ 4 \\ a+b \end{bmatrix}$, $\beta = \begin{bmatrix} 1 \\ 1 \\ b+3 \\ 5 \end{bmatrix}$, 当 a, b 取何值时:

(1) β 不能由 $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ 线性表示?

(2) β 有 $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ 的唯一线性表达式? 写出表达式

解: 设有一组数 x_1, x_2, x_3, x_4 , 使 $(\alpha_1, \alpha_2, \alpha_3, \alpha_4) \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \beta = \begin{bmatrix} 1 \\ 1 \\ b+3 \\ 5 \end{bmatrix}$

则有 $\begin{cases} x_1 + x_2 + x_3 + x_4 = 1 \\ x_2 - x_3 + 2x_4 = 1 \\ 2x_1 + 3x_2 + (a+2)x_3 + 4x_4 = b+3 \\ 3x_1 + 5x_2 + x_3 + (a+b)x_4 = 5 \end{cases}$

增广矩阵 $(A, \beta) = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 2 & 1 \\ 2 & 3 & a+2 & 4 & b+3 \\ 3 & 5 & 1 & a+b & 5 \end{bmatrix}$

$\xrightarrow{r_2-2r_1, r_3-3r_1, r_4-3r_1} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 2 & 1 \\ 0 & 1 & a & 2 & b+1 \\ 0 & 2 & -2 & a+b-2 & 2 \end{bmatrix} \xrightarrow{r_3-r_2, r_4-2r_2} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 2 & 1 \\ 0 & 0 & a+1 & 0 & b \\ 0 & 0 & 0 & a+1 & 0 \end{bmatrix}$

要使 $R(A) < R(A, \beta)$

若 $R(A) = 2$, 则 $a = -1$ 当 $b \neq 0$ 时, $R(A, \beta) = 3$ 满足

故 $a = -1$ 且 $b \neq 0$ 时

要使 $R(A) = R(A, \beta) = 4$.

则 $a \neq -1$, b 为任何值时满足

(3) β 能由 $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ 线性表示, 且表示方法不唯一?

解: 要使 $R(A) = R(A, \beta) < 4$

则 $R(A) = R(A, \beta) = 2$

则 $a = -1$ 且 $b = 0$

$(A, \beta) \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{r_1-r_2, r_3-r_2, r_4-r_2} \begin{bmatrix} 1 & 0 & 2 & -1 & 0 \\ 0 & 1 & -1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\begin{cases} x_1 + 2x_3 - x_4 = 0 \\ x_2 - x_3 + 2x_4 = 1 \end{cases}$ 令 $x_3 = C_1, x_4 = C_2$ ($C_1, C_2 \in \mathbb{R}$)

则 $x_1 = -2C_1 + C_2, x_2 = C_1 - 2C_2 + 1$

此时 $\beta = (-2C_1 + C_2)\alpha_1 + (C_1 - 2C_2 + 1)\alpha_2 + C_1\alpha_3 + C_2\alpha_4$ ($C_1, C_2 \in \mathbb{R}$)

6. 设有向量 $\alpha_1 = \begin{bmatrix} 1 \\ 4 \\ 0 \\ 2 \end{bmatrix}$, $\alpha_2 = \begin{bmatrix} 2 \\ 7 \\ 1 \\ 3 \end{bmatrix}$, $\alpha_3 = \begin{bmatrix} 0 \\ 1 \\ -1 \\ a \end{bmatrix}$, $\beta = \begin{bmatrix} 3 \\ 10 \\ b \\ 4 \end{bmatrix}$, 试问当 a, b 取何值时:

(1) β 不能由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示?

解: 设有一组数 x_1, x_2, x_3 使 $(x_1, x_2, x_3) \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \beta = \begin{bmatrix} 3 \\ 10 \\ b \\ 4 \end{bmatrix}$

$$(\alpha_1, \alpha_2, \alpha_3, \beta) = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 4 & 7 & 1 & 10 \\ 0 & 1 & -1 & b \\ 2 & 3 & a & 4 \end{bmatrix} \xrightarrow[r_4 - 2r_1]{r_2 - 4r_1} \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & -1 & 1 & -2 \\ 0 & 1 & -1 & b \\ 0 & -1 & a & -2 \end{bmatrix}$$

$$\xrightarrow[r_4 + r_2]{r_3 + r_2} \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & -1 & 1 & -2 \\ 0 & 0 & 0 & b-2 \\ 0 & 0 & a-1 & 0 \end{bmatrix} \xrightarrow[r_3 \leftrightarrow r_4]{r_3 \leftrightarrow r_4} \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & -1 & 1 & -2 \\ 0 & 0 & a-1 & 0 \\ 0 & 0 & 0 & b-2 \end{bmatrix}$$

要使 $R(\alpha_1, \alpha_2, \alpha_3) < R(\alpha_1, \alpha_2, \alpha_3, \beta)$,

则 $\begin{cases} a=1 \text{ 且 } b \neq 2 \\ a \neq 1 \text{ 且 } b \neq 2 \end{cases} \Rightarrow b \neq 2$

(2) β 可由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示? 写出表达式

解: $R(\alpha_1, \alpha_2, \alpha_3) = R(\alpha_1, \alpha_2, \alpha_3, \beta)$
 $R(A) = R(A, \beta) = 2 < 3: a=1 \text{ 且 } b=2$
 $(\alpha_1, \alpha_2, \alpha_3, \beta) \rightarrow \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & -1 & 1 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[r_2 \times (-1)]{r_1 + 2r_2} \begin{bmatrix} 1 & 0 & 2 & -1 \\ 0 & -1 & 1 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$
 $\begin{cases} x_1 + 2x_3 = -1 \\ x_2 - x_3 = -2 \end{cases} \text{ 令 } x_3 = c (c \in \mathbb{R}) \text{ 则 } \begin{cases} x_1 = -2c-1 \\ x_2 = c-2 \\ x_3 = c \end{cases} \therefore \beta = (-2c-1)\alpha_1 + (c-2)\alpha_2 + c\alpha_3$
 $R(A) = R(A, \beta) = 3: a \neq 1 \text{ 且 } b=2$
 可唯一线性表示

$(\alpha_1, \alpha_2, \alpha_3, \beta) \rightarrow \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & -1 & 1 & -2 \\ 0 & 0 & a-1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[r_2 \times (-1)]{r_1 + 2r_2} \begin{bmatrix} 1 & 0 & 2 & -1 \\ 0 & -1 & 1 & -2 \\ 0 & 0 & a-1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[r_3 \times (-1)]{r_2 \times (-1)} \begin{bmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 1-a & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$
 $\begin{cases} x_1 + 2x_3 = -1 \\ x_2 - x_3 = 2 \\ (a-1)x_3 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = -1 \\ x_2 = 2 \\ x_3 = 0 \end{cases} \therefore \beta = -\alpha_1 + 2\alpha_2$

7. 设有向量 $\alpha_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $\alpha_2 = \begin{bmatrix} 5 \\ 3 \\ 2 \end{bmatrix}$, $\alpha_3 = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$, $\alpha_4 = \begin{bmatrix} -2 \\ 2 \\ -3 \end{bmatrix}$, A 是三阶矩阵,

且有 $A\alpha_1 = \alpha_2, A\alpha_2 = \alpha_3, A\alpha_3 = \alpha_4$, 试求 $A\alpha_4$

解: $(\alpha_1, \alpha_2, \alpha_3, \alpha_4) = \begin{bmatrix} 1 & 5 & 1 & -2 \\ 1 & 3 & 3 & 2 \\ 0 & 2 & -1 & -3 \end{bmatrix} \xrightarrow[r_2 \leftrightarrow r_3]{r_1 - r_2} \begin{bmatrix} 0 & 2 & -2 & -4 \\ 1 & 3 & 3 & 2 \\ 0 & 2 & -1 & -3 \end{bmatrix} \xrightarrow[r_1 - r_3]{r_2 - r_3} \begin{bmatrix} 0 & 0 & -1 & -1 \\ 1 & 3 & 3 & 2 \\ 0 & 2 & -1 & -3 \end{bmatrix}$
 $\xrightarrow[r_2 \leftrightarrow r_3]{r_1 \leftrightarrow r_2} \begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 2 & -1 & -3 \\ 0 & 0 & -1 & -1 \end{bmatrix} \xrightarrow[r_2 \times (-1)]{r_3 \times (-1)} \begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 2 & -1 & -3 \\ 0 & 0 & 1 & 1 \end{bmatrix} \xrightarrow[r_2 \times \frac{1}{2}]{r_1 - 3r_2} \begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \xrightarrow[r_1 - 3r_2]{r_1 - 3r_3} \begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$
 则有 $\alpha_4 = 2\alpha_1 - \alpha_2 + \alpha_3$. $A\alpha_4 = A(2\alpha_1 - \alpha_2 + \alpha_3)$
 $= 2A\alpha_1 - A\alpha_2 + A\alpha_3$
 $= 2\alpha_2 - \alpha_3 + \alpha_4 = \begin{bmatrix} 7 \\ 5 \\ 2 \end{bmatrix}$

8. 判断下列向量组线性相关还是线性无关

(1) $\alpha_1 = (1, 0, -1)^T$, $\alpha_2 = (-2, 2, 0)^T$, $\alpha_3 = (3, -5, 2)^T$

解: $\because A = (\alpha_1, \alpha_2, \alpha_3) = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 2 & -5 \\ -1 & 0 & 2 \end{bmatrix} \xrightarrow[r_3 + r_1]{r_2 + r_1} \begin{bmatrix} 1 & -2 & 3 \\ 0 & 2 & -5 \\ 0 & -2 & 5 \end{bmatrix} \xrightarrow[r_3 + r_2]{r_3 + r_2} \begin{bmatrix} 1 & -2 & 3 \\ 0 & 2 & -5 \\ 0 & 0 & 0 \end{bmatrix}$ $R(A) = 2 < 3 \therefore$ 线性相关

(2) $\alpha_1 = (1, 1, 3, 1)^T$, $\alpha_2 = (3, -1, 2, 4)^T$, $\alpha_3 = (2, 2, 7, -1)^T$

解: $\because A = (\alpha_1, \alpha_2, \alpha_3) = \begin{bmatrix} 1 & 3 & 2 \\ 1 & -1 & 2 \\ 3 & 2 & 7 \\ 1 & 4 & -1 \end{bmatrix} \xrightarrow[r_3 - 3r_1]{r_2 - r_1} \begin{bmatrix} 1 & 3 & 2 \\ 0 & -4 & 0 \\ 0 & -7 & 1 \\ 0 & 1 & -3 \end{bmatrix} \xrightarrow[r_2 \leftrightarrow r_4]{r_3 + 7r_4} \begin{bmatrix} 1 & 3 & 2 \\ 0 & -4 & 0 \\ 0 & 1 & -23 \\ 0 & 1 & -3 \end{bmatrix} \xrightarrow[r_4 - r_3]{r_2 \times (-1/4)} \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 0 \\ 0 & 1 & -23 \\ 0 & 1 & -3 \end{bmatrix} \xrightarrow[r_4 - r_3]{r_4 - r_3} \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & -20 \\ 0 & 0 & 20 \end{bmatrix}$ $R(A^T) = 3 \therefore$ 线性无关

(3) $\alpha_1 = (1, 0, 0, 2, 5)^T$, $\alpha_2 = (0, 1, 0, 3, 4)^T$, $\alpha_3 = (0, 0, 1, 4, 7)^T$, $\alpha_4 = (2, -3, 4, 11, 12)^T$

解: $\because A = (\alpha_1, \alpha_2, \alpha_3, \alpha_4) = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 4 \\ 2 & 3 & 4 & 11 \\ 5 & 4 & 7 & 12 \end{bmatrix} \xrightarrow[r_4 - 2r_1]{r_5 - 5r_1} \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 4 \\ 0 & 3 & 4 & 7 \\ 0 & 4 & 7 & -3 \end{bmatrix} \xrightarrow[r_5 - 4r_3]{r_4 - 3r_2} \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 4 \\ 0 & 3 & 1 & 19 \\ 0 & 4 & -1 & -19 \end{bmatrix} \xrightarrow[r_5 + r_3]{r_4 - r_3} \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 4 \\ 0 & 3 & 0 & 15 \\ 0 & 4 & 0 & -15 \end{bmatrix} \xrightarrow[r_5 - 4r_3]{r_4 - 3r_2} \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 4 \\ 0 & 3 & 0 & 15 \\ 0 & 4 & 0 & -15 \end{bmatrix} \xrightarrow[r_5 - 4r_3]{r_4 - 3r_2} \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 4 \\ 0 & 3 & 0 & 15 \\ 0 & 4 & 0 & -15 \end{bmatrix}$ $R(A^T) = 4 = R(A) = 4 \therefore$ 线性无关

9. 问 a 取什么值时下列向量组线性相关:

$$\alpha_1 = \begin{bmatrix} a \\ 1 \\ 1 \end{bmatrix}, \alpha_2 = \begin{bmatrix} 1 \\ a \\ -1 \end{bmatrix}, \alpha_3 = \begin{bmatrix} 1 \\ -1 \\ a \end{bmatrix}$$

解: 令 $A = (\alpha_1, \alpha_2, \alpha_3) = \begin{bmatrix} a & 1 & 1 \\ 1 & a & -1 \\ 1 & -1 & a \end{bmatrix} \xrightarrow{r_2-r_3} \begin{bmatrix} a & 1 & 1 \\ 0 & a+1 & -1-a \\ 1 & -1 & a \end{bmatrix} \xrightarrow{r_1-ar_3} \begin{bmatrix} 0 & 1+a & 1-a^2 \\ 0 & a+1 & -1-a \\ 1 & -1 & a \end{bmatrix} \xrightarrow{r_1-r_2} \begin{bmatrix} 0 & 0 & -(a-2)(a+1) \\ 0 & a+1 & -1-a \\ 1 & -1 & a \end{bmatrix}$

$$\xrightarrow{r_2 \div (a+1)} \begin{bmatrix} 0 & 0 & -(a-2)(a+1) \\ 0 & 1 & -1 \\ 1 & -1 & a \end{bmatrix} \xrightarrow{r_1 \leftrightarrow r_3} \begin{bmatrix} 1 & -1 & a \\ 0 & 1 & -1 \\ 0 & 0 & -(a-2)(a+1) \end{bmatrix}$$

要使线性相关, 则 $R(A) < 3$
则 $a=2$ 或 $a=-1$

10. 设向量组 $\alpha_1 = (6, k+1, 3)^T$, $\alpha_2 = (k, 2, -2)^T$, $\alpha_3 = (k, 1, 0)^T$

直接看, 判断 k 取不取 -4 即可

(1) k 为何值时, α_1 与 α_2 线性相关? 线性无关?

解: 令 $A = (\alpha_1, \alpha_2) = \begin{bmatrix} 6 & k \\ k+1 & 2 \\ 3 & -2 \end{bmatrix}$ $A^T = \begin{bmatrix} 6 & k+1 & 3 \\ k & 2 & -2 \end{bmatrix} \xrightarrow{r_2 - \frac{k}{6}r_1} \begin{bmatrix} 6 & k+1 & 3 \\ 0 & -\frac{1}{6}k^2 - \frac{1}{6}k + 2 & -\frac{k}{3} - 2 \end{bmatrix}$

要使线性相关, 则 $R(A) < 2$ 即 $R(A) = R(A^T) = 1$, 则 $6(-\frac{1}{6}k^2 - \frac{1}{6}k + 2) = 0$ 且 $6(-\frac{k}{3} - 2) = 0$ 且 $(k+1)(-\frac{k}{3} - 2) - 3(-\frac{1}{6}k^2 - \frac{1}{6}k + 2) = 0$
得 $k = -4$ 或 $k = 3$, 且 $k = -4$ 故 $k = -4$ 时, 线性相关.

要使线性无关, 则 $R(A) = R(A^T) = 2$, 则 $6(-\frac{1}{6}k^2 - \frac{1}{6}k + 2) \neq 0$ 或 $6(-\frac{k}{3} - 2) \neq 0$ 或 $(k+1)(-\frac{k}{3} - 2) - 3(-\frac{1}{6}k^2 - \frac{1}{6}k + 2) \neq 0$

(2) k 为何值时, $\alpha_1, \alpha_2, \alpha_3$ 线性相关? 线性无关?

得 $k \neq -4$ 且 $k \neq 3$ 或 $k \neq -4$ 或 $k \neq 3$.

解: 由于有 3 个 3 维向量

故当 $k \neq -4$ 时, 线性无关.

可直接由行列式 $\begin{vmatrix} 6 & k & k \\ k+1 & 2 & 1 \\ 3 & -2 & 0 \end{vmatrix} = 3 \times (-1)^{3+1} \begin{vmatrix} k & k \\ 2 & 1 \end{vmatrix} + (-2) \times (-1)^{3+2} \begin{vmatrix} 6 & k \\ k+1 & 1 \end{vmatrix} = -3k + 2(6 - k^2 - k) = -2k^2 - 5k + 12$

$= (-2k+3)(k+4)$ 当其 $= 0$ 时, 即 $k = \frac{3}{2}$ 或 $k = -4$ 时, 线性相关

当 $k \neq \frac{3}{2}$ 且 $k \neq -4$ 时, 线性无关.

(3) 当 $\alpha_1, \alpha_2, \alpha_3$ 线性相关时, 将 α_3 用 α_1, α_2 线性表示

解: ① 当 $k = \frac{3}{2}$ 时, $(\alpha_1, \alpha_2, \alpha_3) = \begin{bmatrix} 6 & \frac{3}{2} & \frac{3}{2} \\ \frac{9}{2} & 2 & 1 \\ 3 & -2 & 0 \end{bmatrix} \xrightarrow[r_1-r_2]{r_2 \times 2} \begin{bmatrix} 1 & -\frac{5}{2} & -\frac{1}{2} \\ 5 & 4 & 2 \\ 3 & -2 & 0 \end{bmatrix} \xrightarrow[r_2-r_1]{r_2 \times \frac{1}{5}} \begin{bmatrix} 1 & -\frac{5}{2} & -\frac{1}{2} \\ 0 & \frac{3}{10} & \frac{9}{10} \\ 3 & -2 & 0 \end{bmatrix} \xrightarrow{r_3-3r_1} \begin{bmatrix} 1 & -\frac{5}{2} & -\frac{1}{2} \\ 0 & \frac{3}{10} & \frac{9}{10} \\ 0 & \frac{11}{2} & \frac{3}{2} \end{bmatrix} \xrightarrow{r_3-\frac{11}{3}r_2} \begin{bmatrix} 1 & -\frac{5}{2} & -\frac{1}{2} \\ 0 & \frac{3}{10} & \frac{9}{10} \\ 0 & 0 & 0 \end{bmatrix}$

$R(\alpha_1, \alpha_2, \alpha_3) = 2$ 而 $R(\alpha_1, \alpha_2) = 2$, 故可唯一线性表示 $\begin{cases} \alpha_2 - \frac{5}{2}\alpha_1 = -\frac{1}{2}\alpha_3 \\ \frac{3}{10}\alpha_2 = \frac{9}{10}\alpha_3 \end{cases} \Rightarrow \begin{cases} \alpha_1 = \frac{2}{3}\alpha_2 \\ \alpha_2 = \frac{3}{11}\alpha_3 \end{cases} \therefore \alpha_3 = \frac{2}{11}\alpha_1 + \frac{3}{11}\alpha_2$

② 当 $k = -4$ 时, $(\alpha_1, \alpha_2, \alpha_3) = \begin{bmatrix} 6 & -4 & -4 \\ -3 & 2 & 1 \\ 3 & -2 & 0 \end{bmatrix} \xrightarrow{r_1+4r_2} \begin{bmatrix} -6 & 4 & 0 \\ -3 & 2 & 1 \\ 3 & -2 & 0 \end{bmatrix} \xrightarrow{r_1+2r_3} \begin{bmatrix} 0 & 0 & 0 \\ -3 & 2 & 1 \\ 3 & -2 & 0 \end{bmatrix} \xrightarrow{r_2+r_3} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 3 & -2 & 0 \end{bmatrix}$

$\xrightarrow[r_1 \leftrightarrow r_3]{r_2 \div 3} \begin{bmatrix} 1 & -\frac{2}{3} & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ $R(\alpha_1, \alpha_2, \alpha_3) = 2$ 而 $R(\alpha_1, \alpha_2) = 1$, 故此时 α_3 不能用 α_1, α_2 线性表示.

11. 设 α_1, α_2 线性无关, $\alpha_1 + \beta$ 与 $\alpha_2 + \beta$ 线性相关, 求向量 β 用 α_1, α_2 线性表示的表示式

解:

$\alpha_1 + \beta$ 与 $\alpha_2 + \beta$ 线性相关

$\therefore$ 存在不全为零的常数 k_1, k_2 , 使 $k_1(\alpha_1 + \beta) + k_2(\alpha_2 + \beta) = 0 \Rightarrow (k_1 + k_2)\beta = -k_1\alpha_1 - k_2\alpha_2$

又 α_1, α_2 线性无关 $\therefore k_1 + k_2 \neq 0$

$$\therefore \beta = -\frac{k_1}{k_1+k_2}\alpha_1 - \frac{k_2}{k_1+k_2}\alpha_2$$

($k_1, k_2 \in \mathbb{R}$ 且 $k_1 + k_2 \neq 0$)

否则 $k_1\alpha_1 + k_2\alpha_2 = 0$, 矛盾

仅 $k_1 = k_2 = 0$ 时成立, 与 $k_1 + k_2 \neq 0$ 矛盾
也与 k_1, k_2 不全为零, 与“行 2”矛盾.

12. 设 α_1, α_2 线性相关, β_1, β_2 线性相关, 问 $\alpha_1 + \beta_1, \alpha_2 + \beta_2$ 是否一定线性相关? 举例说明解:

否

例如: 令 $\alpha_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \alpha_2 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \beta_1 = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \beta_2 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$

则 $\alpha_1 + \beta_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \alpha_2 + \beta_2 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, 此时 $\alpha_1 + \beta_1, \alpha_2 + \beta_2$ 线性无关

13. 设 $\beta_1 = \alpha_1 + \alpha_2, \beta_2 = \alpha_2 + \alpha_3, \beta_3 = \alpha_3 + \alpha_4, \beta_4 = \alpha_4 + \alpha_1$, 证明向量组 $\beta_1, \beta_2, \beta_3, \beta_4$ 线性相关

解:

$\therefore \beta_1 = \alpha_1 + \alpha_2, \beta_2 = \alpha_2 + \alpha_3, \beta_3 = \alpha_3 + \alpha_4, \beta_4 = \alpha_4 + \alpha_1$

$\therefore \alpha_1 = \beta_1 - \alpha_2, \alpha_2 = \beta_2 - \alpha_3, \alpha_3 = \beta_3 - \alpha_4, \alpha_4 = \beta_4 - \alpha_1$

$\therefore \alpha_1 = \beta_1 - (\beta_2 - \alpha_3) = \beta_1 - (\beta_2 - (\beta_3 - \alpha_4)) = \beta_1 - (\beta_2 - (\beta_3 - (\beta_4 - \alpha_1)))$
 $= \beta_1 - \beta_2 + \beta_3 - \beta_4 + \alpha_1$

$\therefore \beta_1 - \beta_2 + \beta_3 - \beta_4 = 0$, 故 $\beta_1, \beta_2, \beta_3, \beta_4$ 线性相关

14. 设 $\beta_1 = \alpha_1, \beta_2 = \alpha_1 + \alpha_2, \dots, \beta_r = \alpha_1 + \alpha_2 + \dots + \alpha_r$, 且向量组 $\alpha_1, \alpha_2, \dots, \alpha_r$ 线性无关, 证明向量组 $\beta_1, \beta_2, \dots, \beta_r$ 线性无关

解:

设 $k_1\beta_1 + k_2\beta_2 + \dots + k_r\beta_r = 0$, 则 $k_1\alpha_1 + k_2(\alpha_1 + \alpha_2) + \dots + k_r(\alpha_1 + \alpha_2 + \dots + \alpha_r) = 0$

即 $(k_1 + \dots + k_r)\alpha_1 + (k_2 + \dots + k_r)\alpha_2 + \dots + k_r\alpha_r = 0$

又 $\alpha_1, \alpha_2, \dots, \alpha_r$ 线性无关 $\therefore \begin{cases} k_1 + \dots + k_r = 0 \\ k_2 + \dots + k_r = 0 \\ \vdots \\ k_r = 0 \end{cases} \Leftrightarrow \begin{bmatrix} 1 & 1 & \dots & 1 \\ 0 & 1 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \\ \vdots \\ k_r \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$

$\therefore \begin{vmatrix} 1 & 1 & \dots & 1 \\ 0 & 1 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{vmatrix} = 1 \neq 0 \therefore$ 方程组只有零解, 即 $k_1 = k_2 = \dots = k_r = 0$

故 $\beta_1, \beta_2, \dots, \beta_r$ 线性无关

15. 设向量组 $\alpha_1, \alpha_2, \alpha_3$ 线性无关, 已知 $\beta_1 = k_1\alpha_1 + \alpha_2 + k_2\alpha_3, \beta_2 = \alpha_1 + k_2\alpha_2 + (k_2+1)\alpha_3, \beta_3 = \alpha_1 + \alpha_2 + \alpha_3$, 试问当 k_1, k_2 为何值时, $\beta_1, \beta_2, \beta_3$ 线性相关? 线性无关?

解:

由题意得

$(\beta_1, \beta_2, \beta_3) = (\alpha_1, \alpha_2, \alpha_3) \begin{bmatrix} k_1 & 1 & 1 \\ 1 & k_2 & 1 \\ k_1 & k_2+1 & 1 \end{bmatrix}$

令 $B = (\beta_1, \beta_2, \beta_3), A = (\alpha_1, \alpha_2, \alpha_3), P = \begin{bmatrix} k_1 & 1 & 1 \\ 1 & k_2 & 1 \\ k_1 & k_2+1 & 1 \end{bmatrix}$

即有 $B = AP$

$\because \alpha_1, \alpha_2, \alpha_3$ 线性无关 $\therefore R(A) = 3$

而 $|P| = \begin{vmatrix} k_1 & 1 & 1 \\ 1 & k_2 & 1 \\ k_1 & k_2+1 & 1 \end{vmatrix} \xrightarrow{r_3-r_1} \begin{vmatrix} k_1 & 1 & 1 \\ 1 & k_2 & 1 \\ 0 & k_2 & 0 \end{vmatrix} \xrightarrow{r_2-r_3} \begin{vmatrix} k_1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & k_2 & 0 \end{vmatrix} = k_2(-1)^{3+2} \cdot (k_1-1) = (1-k_1)k_2$

故当 $k_1 \neq 1$ 且 $k_2 \neq 0$ 时, $R(B) = 3, \beta_1, \beta_2, \beta_3$ 线性无关

当 $k_1 = 1$ 或 $k_2 = 0$ 时, $R(B) < 3, \beta_1, \beta_2, \beta_3$ 线性相关

16. 设向量组 $\alpha_1, \alpha_2, \dots, \alpha_s$ 线性相关, 且其中任意 $(s-1)$ 个向量都线性无关, 试证明: 必然存在一组全都不为零的数 $k_1, k_2, \dots, k_s$, 使 $k_1\alpha_1 + k_2\alpha_2 + \dots + k_s\alpha_s = 0$.

证明:

$\because \alpha_1, \alpha_2, \dots, \alpha_s$ 线性相关.

$\therefore$ 存在不全为零的数 $k_1, k_2, \dots, k_s$, 使 $k_1\alpha_1 + k_2\alpha_2 + \dots + k_s\alpha_s = 0$

① 当 $k_1, k_2, \dots, k_s$ 全都不为零, 得证

② 当 $k_1, k_2, \dots, k_s$ 中至少有一个为零. 不妨假设 $k_1 = 0, k_2, \dots, k_s$ 全都不为零时,

$$\text{使 } (k_1\alpha_1) + k_2\alpha_2 + \dots + k_s\alpha_s = 0$$

由 $k_2\alpha_2 + \dots + k_s\alpha_s = 0$ 得 $\alpha_2, \alpha_3, \dots, \alpha_s$ 线性相关, 矛盾

17. 设三维列向量 $\alpha_1, \alpha_2, \alpha_3$ 线性无关, A 是三阶矩阵, 且有

$$A\alpha_1 = \alpha_1 + 2\alpha_2 + 3\alpha_3, \text{ 求 } |A|$$

$$A\alpha_2 = 2\alpha_2 + 3\alpha_3$$

$$A\alpha_3 = 3\alpha_2 - 4\alpha_3$$

解:

$$\text{由题意得: } A(\alpha_1, \alpha_2, \alpha_3) = (A\alpha_1, A\alpha_2, A\alpha_3) = (\alpha_1, \alpha_2, \alpha_3) \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 3 \\ 3 & 3 & -4 \end{bmatrix}$$

$\rightarrow R(A) = 3$, 满秩

$\therefore \alpha_1, \alpha_2, \alpha_3$ 线性无关 $\therefore |\alpha_1, \alpha_2, \alpha_3| \neq 0$

行列式不为0的矩阵一定可逆

$$\therefore |A(\alpha_1, \alpha_2, \alpha_3)| = |A| |\alpha_1, \alpha_2, \alpha_3| = |\alpha_1, \alpha_2, \alpha_3| \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 3 \\ 3 & 3 & -4 \end{bmatrix} = |\alpha_1, \alpha_2, \alpha_3| \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 3 \\ 3 & 3 & -4 \end{bmatrix}$$

$$\therefore |A| = \begin{vmatrix} 1 & 0 & 0 \\ 2 & 2 & 3 \\ 3 & 3 & -4 \end{vmatrix} = -1$$

18. 设向量组 $A: \alpha_1 = (1, 2, 1, 3)^T, \alpha_2 = (4, -1, -5, -6)^T$

向量组 $B: \beta_1 = (-1, 3, 4, 7)^T, \beta_2 = (2, -1, -3, -4)^T$

试证明: 向量组 A 与向量组 B 等价

证明:

$$\therefore (\alpha_1, \alpha_2, \beta_1, \beta_2) = \begin{bmatrix} 1 & 4 & -1 & 2 \\ 2 & -1 & 3 & -1 \\ 1 & -5 & 4 & -3 \\ 3 & -6 & 7 & -4 \end{bmatrix} \xrightarrow[r_4 - 3r_1]{r_2 - 2r_1, r_3 - r_1} \begin{bmatrix} 1 & 4 & -1 & 2 \\ 0 & -9 & 5 & -5 \\ 0 & -9 & 5 & -5 \\ 0 & -18 & 10 & -10 \end{bmatrix} \xrightarrow[r_2 \div (-9), r_4 \div (-9)]{r_3 - r_2} \begin{bmatrix} 1 & 4 & -1 & 2 \\ 0 & 1 & -\frac{5}{9} & \frac{5}{9} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ 向量组 A 与 B 秩相同, 故二者等价

$$19. \text{ 设 } \begin{cases} \beta_1 = \alpha_2 + \alpha_3 + \dots + \alpha_n \\ \beta_2 = \alpha_1 + \alpha_3 + \dots + \alpha_n \\ \dots \\ \beta_n = \alpha_1 + \alpha_2 + \dots + \alpha_{n-1} \end{cases}$$

证明: 向量组 A 与向量组 B 等价

$\alpha_1, \alpha_2, \dots, \alpha_n$

$\beta_1, \beta_2, \dots, \beta_n$

证明:

$$\therefore (\beta_1, \beta_2, \dots, \beta_n) = (\alpha_1, \alpha_2, \dots, \alpha_n) K$$

$$\therefore |K| = \begin{vmatrix} 0 & 1 & \dots & 1 \\ 1 & 0 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \dots & 0 \end{vmatrix} \xrightarrow{T_1 + T_2 + \dots + T_n} \begin{vmatrix} 0 & 1 & \dots & 1 \\ 1 & 0 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \dots & 0 \end{vmatrix} \xrightarrow{C_m - C_1 (m=2, \dots, n)} \begin{vmatrix} 0 & 1 & \dots & 0 \\ 1 & -1 & \dots & 0 \\ \vdots & 0 & \ddots & 0 \\ 1 & 0 & \dots & -1 \end{vmatrix}$$

$$= (n-1) \cdot (-1)^{n-1} \neq 0$$

$\therefore K$ 可逆

故由 $B = AK$ 得 $A = BK^{-1}$

★

A 能由 B 线性表示且系数矩阵为 K^{-1}

故二者等价.

20. 设有两个向量组 $\alpha_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \\ 3 \end{bmatrix}$, $\alpha_2 = \begin{bmatrix} 2 \\ 5 \\ 9 \\ 8 \end{bmatrix}$, $\alpha_3 = \begin{bmatrix} -1 \\ 0 \\ 3 \\ 7 \end{bmatrix}$,

$\beta_1 = \begin{bmatrix} 1 \\ a \\ a+5 \\ 7 \end{bmatrix}$, $\beta_2 = \begin{bmatrix} 3 \\ 3+a \\ 3 \\ 11 \end{bmatrix}$, $\beta_3 = \begin{bmatrix} 0 \\ 1 \\ 6 \\ 2 \end{bmatrix}$,

若 β_1 可由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示, 试判断这两个向量组是否等价?

解.

$\because \beta_1$ 可由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示

$$\therefore (\alpha_1, \alpha_2, \alpha_3, \beta_1) = \begin{bmatrix} 1 & 2 & -1 & 1 \\ 2 & 5 & 0 & a \\ -1 & a & 3 & a+5 \\ 3 & 8 & 1 & 7 \end{bmatrix} \xrightarrow[r_4 \rightarrow r_1]{\substack{r_2 - 2r_1 \\ r_3 + r_1 \\ r_4 - 3r_1}} \begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & 2 & a-2 \\ 0 & a+2 & 2 & a-4 \\ 0 & 2 & 4 & 4 \end{bmatrix}$$

$$\xrightarrow[r_4 - 2r_2]{\substack{r_3 - (a+2)r_2 \\ r_4 - 2r_2}} \begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & 2 & a-2 \\ 0 & 0 & -2a-2 & 0 \\ 0 & 0 & 0 & 8-2a \end{bmatrix} \xrightarrow[r_4 \div 2]{r_3 \div (-2)} \begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & 2 & a-2 \\ 0 & 0 & a+1 & 0 \\ 0 & 0 & 0 & 4-a \end{bmatrix}$$

$R(\alpha_1, \alpha_2, \alpha_3) = R(\alpha_1, \alpha_2, \alpha_3, \beta_1)$ 得 $a=4$

$$\therefore (\alpha_1, \alpha_2, \alpha_3, \beta_1) \rightarrow \begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[r_3 \div 5]{r_1 - 2r_2} \begin{bmatrix} 1 & 0 & -9 & -3 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow[r_2 - 2r_3]{r_1 + 9r_3} \begin{bmatrix} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{cases} \alpha_1 = -3 \\ \alpha_2 = 2 \\ \alpha_3 = 0 \end{cases} \Rightarrow \beta_1 = -3\alpha_1 + 2\alpha_2$$

$\therefore a=4 \quad \therefore \beta_2 = \begin{bmatrix} 3 \\ 7 \\ 3 \\ 11 \end{bmatrix}, \beta_3 = \begin{bmatrix} 0 \\ 1 \\ 6 \\ 2 \end{bmatrix}$

$= \alpha_1 + \alpha_2 \qquad \qquad \qquad = \alpha_2 - 2\alpha_1$

$\therefore$ 向量组 $\beta_1, \beta_2, \beta_3$ 可由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示.

而 $\beta_1, \beta_2, \beta_3$ 仅可由 α_1, α_2 线性表示且 α_1, α_2 线性无关

$\therefore R(\alpha_1, \alpha_2) = 2 = n.$

$\therefore R(\alpha_1, \alpha_2) = R(\alpha_1, \alpha_2, \beta_1, \beta_2, \beta_3) = 2$, 而 $R(\alpha_1, \alpha_2, \alpha_3) = 3$

$R(\beta_1, \beta_2, \beta_3) \neq R(\alpha_1, \alpha_2, \alpha_3)$

$\therefore$ 这两个向量组不等价

向量组的秩

对向量组 A , 若能在 A 中选出 r 个向量 $\alpha_1, \alpha_2, \dots, \alpha_r$ 满足

- (1) 向量组 $A_0: \alpha_1, \alpha_2, \dots, \alpha_r$ 线性无关
- (2) A 中任意 $r+1$ 个向量(若有的话)都线性相关

则称向量组 $\alpha_1, \alpha_2, \dots, \alpha_r$ 为向量组 A 的最大(线性)无关(向量)组,
最大无关组所含的向量个数 r 称为 A 的秩, 记为 $R(A)$

只有零向量的向量组没有最大无关组, 规定其秩为 0

例如, 对二维向量组 $\alpha_1 = (0, 1)^T, \alpha_2 = (1, 0)^T, \alpha_3 = (1, 1)^T, \alpha_4 = (0, 2)^T$

$\therefore \alpha_1, \alpha_2, \alpha_3 \quad \alpha_1, \alpha_2, \alpha_4 \quad \alpha_1, \alpha_3, \alpha_4 \quad \alpha_2, \alpha_3, \alpha_4$ 都线性相关
 $\alpha_3 = \alpha_1 + \alpha_2 \quad \alpha_4 = 2\alpha_1 + 0 \cdot \alpha_2 \quad \alpha_4 = 2\alpha_1 + 0 \cdot \alpha_3 \quad \alpha_2 = \alpha_3 - \frac{1}{2}\alpha_4$

$\therefore \alpha_1, \alpha_2$ 线性无关 $\therefore$ 它们都是最大无关组
 α_2, α_3

由: $\alpha_1, \alpha_2, \dots, \alpha_m$ 线性无关而 $\alpha_1, \alpha_2, \dots, \alpha_m, \beta$ 线性相关, 则 β 可由 $\alpha_1, \alpha_2, \dots, \alpha_m$ 唯一线性表示

知: 当 A 中 r 个向量 $\alpha_1, \alpha_2, \dots, \alpha_r$ 线性无关时, 条件1

A 中任意 $r+1$ 个向量都线性相关 $\Leftrightarrow A$ 中任一向量都可用 $\alpha_1, \alpha_2, \dots, \alpha_r$ 线性表示

条件2

若 $\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m \end{cases}$ 对应的行向量组 $\alpha_1, \alpha_2, \dots, \alpha_m$

其秩就是其中线性无关方程组的个数, 为 r

$\rightarrow$ 就是由它们组成同解方程组, 其余 $(m-r)$ 个都多余了

- 向量组 A 与其最大无关组 A_0 等价
- 矩阵的秩等于它的列(行)向量组的秩
- 矩阵的行向量组和列向量组的秩是相等的
- 等价的向量组秩相等

求最大无关组的方法:

以向量组中各向量为列向量组成矩阵 A , 用初等行变换把 A 化为行阶梯形矩阵 B

假设秩为 r , 从 B 中选取一个不为零的 r 阶子式, 这个 r 阶子式包含的 r 个列向量就是一个最大无关组

例. 求下列向量组的秩, 并给出一个最大无关组.

$$\alpha_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \\ 1 \end{bmatrix}, \quad \alpha_2 = \begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \quad \alpha_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \quad \alpha_4 = \begin{bmatrix} 1 \\ 1 \\ 3 \\ -1 \\ 3 \end{bmatrix}$$

解:

$$A = (\alpha_1, \alpha_2, \alpha_3, \alpha_4) = \begin{bmatrix} 1 & -1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 3 \\ -1 & 0 & 0 & -1 \\ 1 & 1 & 1 & 3 \end{bmatrix}$$

$$\xrightarrow{\substack{r_3-r_1 \\ r_4+r_1 \\ r_5-r_1}} \begin{bmatrix} 1 & -1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 2 & 0 & 2 \\ 0 & -1 & 1 & 0 \\ 0 & 2 & 0 & 2 \end{bmatrix} \xrightarrow{\substack{r_5-r_3 \\ r_3-2r_2}} \begin{bmatrix} 1 & -1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\substack{r_5 \leftrightarrow r_4 \\ r_3+r_2}} \begin{bmatrix} 1 & -1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = B$$

$\therefore R(A) = 3$, 即向量组秩为3.

故在B中选取一个不为零的3阶子式:

$\therefore \alpha_1, \alpha_2, \alpha_3$ 这3个列向量对应的一个3阶子式不为0

$\therefore \alpha_1, \alpha_2, \alpha_3$ 是向量组的一个最大无关组

(同理, $\alpha_1, \alpha_2, \alpha_4$ 也是)

例. 求向量组 $\alpha_1 = (1, 2, -1, 1)^T$, $\alpha_2 = (2, 0, t, 0)^T$, $\alpha_3 = (0, -4, 5, -2)^T$, $\alpha_4 = (3, -2, t+4, -1)^T$ 的秩和最大无关组

解: $A = (\alpha_1, \alpha_2, \alpha_3, \alpha_4) = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 0 & -4 & -2 \\ -1 & t & 5 & t+4 \\ 1 & 0 & -2 & -1 \end{bmatrix} \xrightarrow{\substack{r_2-2r_1 \\ r_3+r_1 \\ r_4-r_1}} \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & -4 & -4 & -8 \\ 0 & t+2 & 5 & t+7 \\ 0 & -2 & -2 & -4 \end{bmatrix} \xrightarrow{\substack{r_4-\frac{1}{2}r_2 \\ r_2 \div (-4)}} \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & t+2 & 5 & t+7 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

(线性无关, 子式也不为零)

当 $t=3$ 时, 向量组秩为2, 且 α_1, α_2 是最大无关组

当 $t \neq 3$ 时, 向量组秩为3, 且 $\alpha_1, \alpha_2, \alpha_3$ 是最大无关组

例. 证明: $R(AB) \leq \min\{R(A), R(B)\}$

证明: 设 $A = (a_{ij})_{m \times n} = (\alpha_1, \alpha_2, \dots, \alpha_n)$, $B = (b_{ij})_{n \times s} = \begin{bmatrix} b_{11} \\ b_{12} \\ \vdots \\ b_{ns} \end{bmatrix}$

则 $AB = C = (c_{ij})_{m \times s} = (\beta_1, \beta_2, \dots, \beta_s)$

$$= (\alpha_1, \alpha_2, \dots, \alpha_n) \begin{bmatrix} b_{11} & b_{12} & \dots & b_{1s} \\ b_{21} & b_{22} & \dots & b_{2s} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \dots & b_{ns} \end{bmatrix}$$

左 右
 $r_i = \text{第1行} \times \text{第1列} + \text{第2行} \times \text{第1列} + \dots + \text{第n行} \times \text{第1列}$
 $= \alpha_1 b_{11} + \alpha_2 b_{21} + \dots + \alpha_n b_{n1}$

$\dots$ 归纳, 得 $r_j = b_{1j}\alpha_1 + b_{2j}\alpha_2 + \dots + b_{nj}\alpha_n \quad (j=1, 2, \dots, s)$

$\therefore AB$ 的列向量组 $\beta_1, \beta_2, \dots, \beta_s$ 可由 A 的列向量组 $\alpha_1, \alpha_2, \dots, \alpha_n$ 线性表示

$\beta_1, \beta_2, \dots, \beta_s$ 的最大无关组可由 $\alpha_1, \alpha_2, \dots, \alpha_n$ 的最大无关组线性表示

$\therefore R(AB) \leq R(A)$ 同理: $R(AB) \leq R(B) \quad \therefore R(AB) \leq \min\{R(A), R(B)\}$

1. 求下列向量组的秩和一个最大无关组, 并将其余向量用此最大无关组线性表示

(1) $\alpha_1 = (1, 1, 1)^T$, $\alpha_2 = (1, 1, 0)^T$, $\alpha_3 = (1, 0, 0)^T$, $\alpha_4 = (1, 2, -3)^T$

解: $A = (\alpha_1, \alpha_2, \alpha_3, \alpha_4) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 2 \\ 1 & 0 & 0 & -3 \end{bmatrix} \xrightarrow[r_2-r_3]{r_1-r_2} \begin{bmatrix} 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & 5 \\ 1 & 0 & 0 & -3 \end{bmatrix} \xrightarrow{r_1 \leftrightarrow r_3} \begin{bmatrix} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & -1 \end{bmatrix}$ 该向量组秩为3

$\alpha_1, \alpha_2, \alpha_3$ 是一个最大无关组 且 $\alpha_4 = -3\alpha_1 + 5\alpha_2 - \alpha_3$

(2) $\alpha_1 = (2, 1, 1, 1)^T$, $\alpha_2 = (-1, 1, 7, 10)^T$, $\alpha_3 = (3, 1, -1, -2)^T$, $\alpha_4 = (8, 5, 9, 11)^T$

解: $A = (\alpha_1, \alpha_2, \alpha_3, \alpha_4) = \begin{bmatrix} 2 & -1 & 3 & 8 \\ 1 & 1 & 1 & 5 \\ 1 & 7 & -1 & 9 \\ 1 & 10 & -2 & 11 \end{bmatrix} \xrightarrow[r_4-r_2]{r_1-2r_2, r_3-r_2} \begin{bmatrix} 0 & -3 & 1 & -2 \\ 1 & 1 & 1 & 5 \\ 0 & 6 & -2 & 4 \\ 0 & 9 & -3 & 6 \end{bmatrix} \xrightarrow[r_3+2r_1]{r_4+3r_1} \begin{bmatrix} 0 & -3 & 1 & -2 \\ 1 & 1 & 1 & 5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[r_2-r_1]{r_1 \div (-3)} \begin{bmatrix} 0 & 1 & -\frac{1}{3} & \frac{2}{3} \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{r_1 \leftrightarrow r_2} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -\frac{1}{3} & \frac{2}{3} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

该向量组秩为2 α_1, α_2 是一个最大无关组, 且 $\alpha_3 = \frac{1}{3}\alpha_1 - \frac{2}{3}\alpha_2$, $\alpha_4 = \frac{13}{3}\alpha_1 + \frac{2}{3}\alpha_2$

(3) $\alpha_1 = (1, 1, 3, 1)^T$, $\alpha_2 = (-1, 1, -1, 3)^T$, $\alpha_3 = (5, -2, 8, -9)^T$, $\alpha_4 = (-1, 3, 1, 7)^T$

解: 对 $A = (\alpha_1, \alpha_2, \alpha_3, \alpha_4)$ 施以初等行变换 $A = \begin{bmatrix} 1 & -1 & 5 & -1 \\ 1 & 1 & -1 & 3 \\ 3 & -1 & 8 & 1 \\ 1 & 3 & -9 & 7 \end{bmatrix} \xrightarrow[r_4-r_1]{r_2-r_1, r_3-3r_1} \begin{bmatrix} 1 & -1 & 5 & -1 \\ 0 & 2 & -6 & 4 \\ 0 & 2 & -7 & 4 \\ 0 & 4 & -14 & 8 \end{bmatrix} \xrightarrow[r_4-2r_2]{r_3-r_2} \begin{bmatrix} 1 & -1 & 5 & -1 \\ 0 & 2 & -6 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[r_1+r_2]{r_2 \div 2} \begin{bmatrix} 1 & 0 & \frac{7}{2} & \frac{3}{2} \\ 0 & 1 & -3 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ 该向量组秩为2, α_1, α_2 是一个最大无关组 且 $\alpha_3 = \frac{7}{2}\alpha_1 - 3\alpha_2$, $\alpha_4 = \alpha_1 + 2\alpha_2$

(4) $\alpha_1 = (1, -1, 0, 4)^T$, $\alpha_2 = (2, 1, 5, 6)^T$, $\alpha_3 = (1, -1, -2, 0)^T$, $\alpha_4 = (3, 0, 7, 14)^T$

解: $A = (\alpha_1, \alpha_2, \alpha_3, \alpha_4) = \begin{bmatrix} 1 & 2 & 1 & 3 \\ -1 & 1 & -1 & 0 \\ 0 & 5 & -2 & 7 \\ 4 & 6 & 0 & 14 \end{bmatrix} \xrightarrow[r_4-4r_1]{r_2+r_1, r_3-5r_2} \begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 3 & 0 & 3 \\ 0 & 5 & -2 & 7 \\ 0 & -2 & -4 & 2 \end{bmatrix} \xrightarrow[r_4+2r_2]{r_2 \div 3} \begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 5 & -2 & 7 \\ 0 & 0 & -4 & 4 \end{bmatrix} \xrightarrow[r_3-5r_2]{r_4 \div (-4)} \begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -2 & 2 \\ 0 & 0 & 1 & -1 \end{bmatrix} \xrightarrow[r_3 \div (-2)]{r_3 \leftrightarrow r_4} \begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{r_1-2r_2} \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{r_1-r_3} \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ 该向量组秩为3. $\alpha_1, \alpha_2, \alpha_3$ 是一个最大无关组 且 $\alpha_4 = 2\alpha_1 + \alpha_2 - \alpha_3$

2. 设向量组 $\alpha_1 = \begin{bmatrix} a \\ 3 \\ 1 \end{bmatrix}$, $\alpha_2 = \begin{bmatrix} 2 \\ b \\ 3 \end{bmatrix}$, $\alpha_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$, $\alpha_4 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$ 的秩为2, 求a, b.

$A = (\alpha_3, \alpha_4, \alpha_1, \alpha_2) = \begin{bmatrix} 1 & 2 & a & 2 \\ 2 & 3 & 3 & b \\ 1 & 1 & 1 & 3 \end{bmatrix} \xrightarrow[r_3-r_1]{r_2-2r_1} \begin{bmatrix} 1 & 2 & a & 2 \\ 0 & -1 & 3-2a & b-4 \\ 0 & -1 & 1-a & 1 \end{bmatrix} \xrightarrow{r_3-r_2} \begin{bmatrix} 1 & 2 & a & 2 \\ 0 & -1 & 3-2a & b-4 \\ 0 & 0 & a-2 & 5-b \end{bmatrix}$

$\therefore R(A) = 2 \quad \therefore a = 2 \text{ 且 } b = 5$

3. 设 $\alpha_1, \alpha_2, \dots, \alpha_n$ 是一组n维向量, 已知n维单位坐标向量 $e_1, e_2, \dots, e_n$ 可由它们线性表示,

证明 $\alpha_1, \alpha_2, \dots, \alpha_n$ 线性无关

证明:

方法二:

$\because e_1, e_2, \dots, e_n$ 可由 $\alpha_1, \alpha_2, \dots, \alpha_n$ 线性表示

$\therefore e_1, e_2, \dots, e_n$ 线性无关

$\therefore n = R(e_1, e_2, \dots, e_n) \leq R(\alpha_1, \alpha_2, \dots, \alpha_n) \leq n$

不妨设 $e_1 = k_{11}\alpha_1 + k_{12}\alpha_2 + \dots + k_{1n}\alpha_n$

$e_2 = k_{21}\alpha_1 + k_{22}\alpha_2 + \dots + k_{2n}\alpha_n$

故 $R(\alpha_1, \alpha_2, \dots, \alpha_n) = n$

$e_n = k_{n1}\alpha_1 + k_{n2}\alpha_2 + \dots + k_{nn}\alpha_n$

$\therefore \alpha_1, \alpha_2, \dots, \alpha_n$ 线性无关

$\begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix} = \begin{bmatrix} k_{11} & k_{12} & \dots & k_{1n} \\ k_{21} & k_{22} & \dots & k_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ k_{n1} & k_{n2} & \dots & k_{nn} \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix}$

两边取行列式.

由 $\begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix} \neq 0$ 得 $\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix} \neq 0$

行列式结果不为0 $\therefore$ 满秩 $\therefore$ 线性无关

$\therefore n$ 维向量组 $\alpha_1, \alpha_2, \dots, \alpha_n$ 构成的矩阵的秩为n. $\therefore \alpha_1, \alpha_2, \dots, \alpha_n$ 线性无关